

Suricata Integration

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1 Introduction

Network security relies on real-time visibility and active defense mechanisms to detect and mitigate unauthorized access, service disruptions, and malicious activity across an enterprise network. **Intrusion Detection Systems (IDS)** and **Intrusion Prevention Systems (IPS)** serve as core security controls designed to analyze network traffic against signature databases and anomaly baselines, ensuring threats are identified before they can compromise internal infrastructure.

2 IDS vs IPS

2.1 IDS

- Inspects copies of network traffic out-of-band without sitting directly in the live data path.
- Logs events and triggers notifications when threats are spotted, without stopping or modifying traffic.
- Operates safely alongside the network, ensuring false positives never block legitimate user traffic.

2.2 IPS

- Sits directly inside the network path, requiring all live traffic to pass through it for inspection.
- Instantly blocks threats by dropping malicious packets, terminating connections, or updating firewall rules.
- Takes action on live traffic, meaning false positives can accidentally block critical legitimate services.

3 What is Suricata?

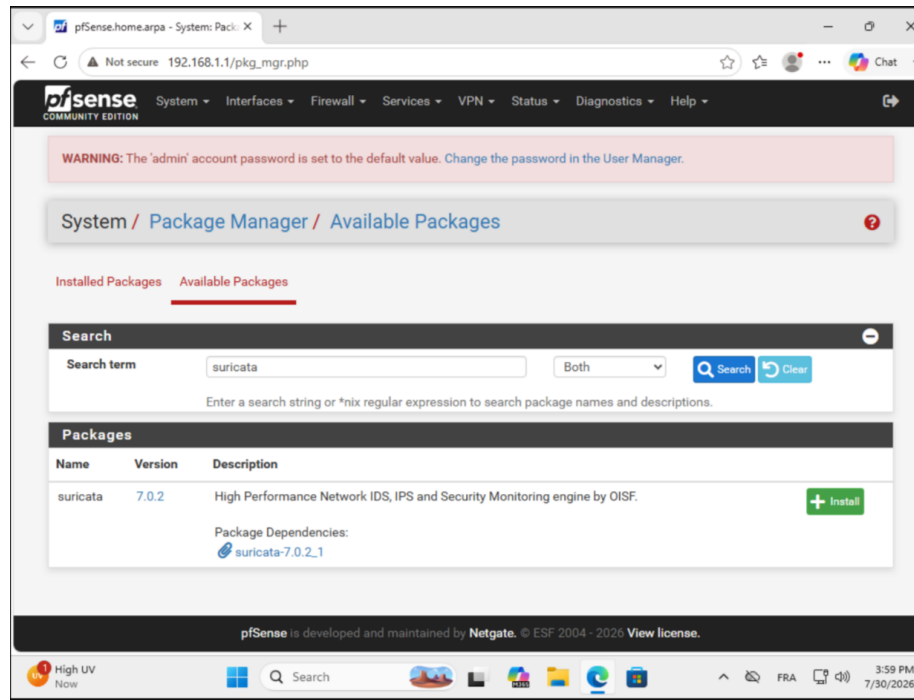
Suricata is a high-performance, open-source network security engine. It functions as an **IDS** (detecting threats), an **IPS** (blocking threats), and an **NSM** (logging detailed network traffic and protocol telemetry). We chose it because its multi-threaded architecture smoothly handles high-speed traffic, and its native JSON logging (`eve.json`) allows seamless integration with SIEM platforms like Wazuh.

4 Suricata Instalation and Configuration

4.1 Instalation

To establish perimeter network threat detection, we deploy Suricata directly on our pfSense gateway to inspect all incoming traffic from the external network before it reaches our internal endpoints.

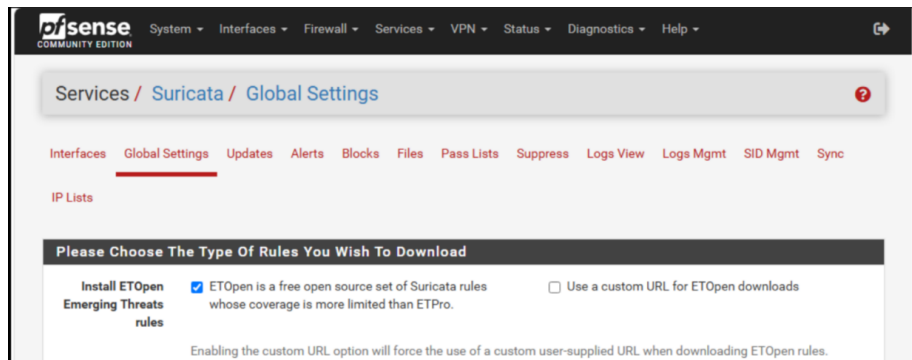
To do that we navigate to **System > Package Manager > Available Packages** within the pfSense WebGUI . We search for **suricata**, then we install the package



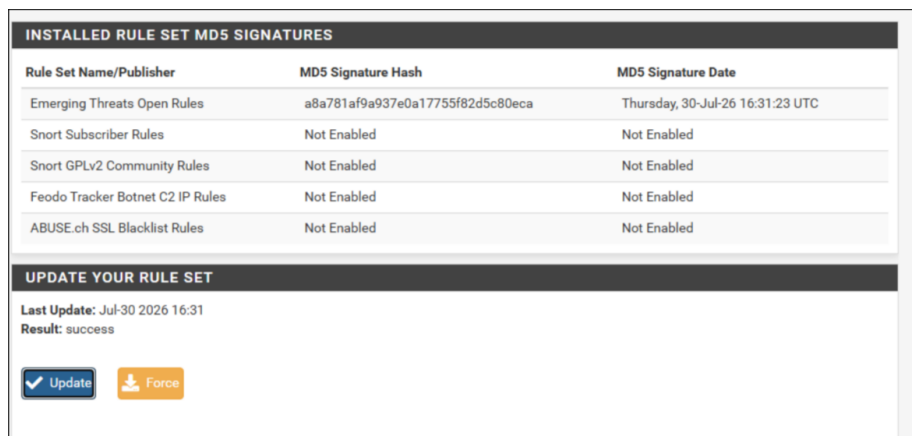
4.2 Rules and Settings Configuration

4.2.1 Threat Signature Configuration

We enable **ET Open Rules** in **Services > Suricata > Global Settings** and save the changes.



We open the **Updates** tab and click **Update Rules** to download the latest threat signatures.



4.2.2 Interface & EVE JSON Logging Setup

We navigate to **Services > Suricata > Interfaces**, add a new instance bound to our **LAN** network interface, and enable **EVE JSON Log** under Logging Settings.

4.2.3 Wazuh Rule Configuration

Since pfSense transmits Suricata alerts via syslog (`program_name: suricata`) rather than JSON, default rules do not trigger. We added a custom rule to `/var/ossec/etc/rules/local_rules.xml` on the Wazuh server to ingest these events:

```
<group name="pfsense,suricata,">

  <!-- Suricata Syslog Alert Rule -->
  <rule id="100120" level="7">
    <match>suricata</match>
    <description>Suricata IDS: Security Alert Detected</description>
    <group>suricata,ids,</group>
  </rule>

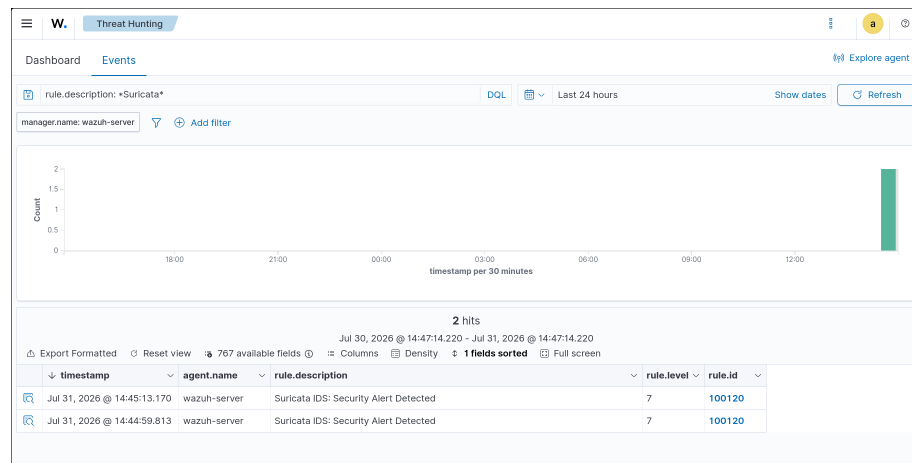
</group>
```

5 Threat Emulation

To validate that the Suricata deployment actively inspects network traffic and streams detection events to Wazuh, we executed a controlled threat emulation test from the Windows endpoint:

```
curl http://testmyids.com
```

When checking the Wazuh Dashboard and filtering by `rule.id: 100120` (or searching for `suricata`), we can see the real-time `testmyids.com` threat alert indexed under Level 7 severity.



Conclusion

Setting up Suricata on pfSense and integrating it with Wazuh creates an effective intrusion detection workflow. By properly configuring syslog forwarding to DAEMON and adding custom SIEM rules, live IDS alerts stream directly to the dashboard for real-time threat monitoring.